RESEARCH DISSERTATION

Characterization of viscoelastic flows in a four-roll mill with physics-informed neural networks

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The Viscoelasticity Problem: Relevance and Mathematical Complexity



01 / FLUID PHYSICS

Elasticity & Fading Memory

Viscous dissipation, elasticity & deformation history

02 / APPLICATIONS

Technological & Bio Systems

Polymer extrusion, spinning, blood & microfluidics

03 / MODELING

Governing Equations

Non-linear constitutive models (Oldroyd-B, PTT) & mixed nature

The 4-Roll Mill: Principles & Characteristics

01 / KINEMATICS

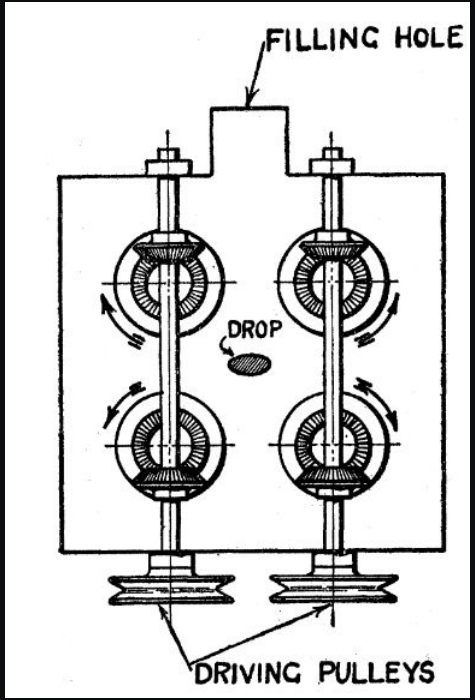
Independent Roll Rotation

Independent roll speed and direction control to generate varied flow regimes

02 / CENTRAL REGION

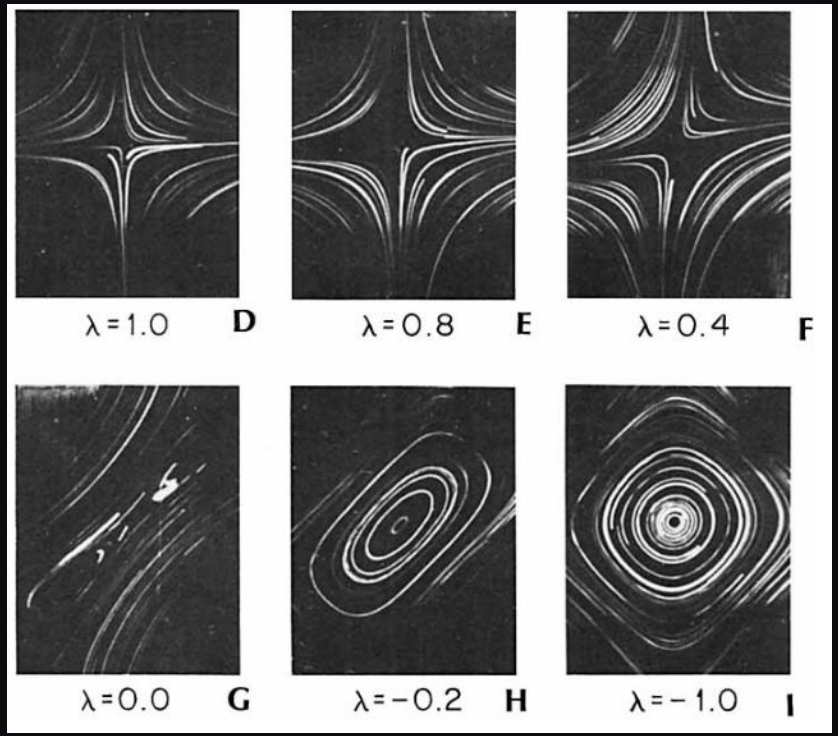
Stagnation Point

Extensional stagnation point created at the center for fluid characterization.



Reference: Taylor et al. (1934) - The formation of emulsions in definable fields of flow
Reference: Fuller et al. (1980) — Flow birefringence of dilute polymer solutions in two-dimensional flows

4-Roll Mill Flow Fields



$$\alpha \equiv \lambda$$

01 / FLOW REGIME

Extensional Flow (α=1)

Generated when opposing rolls rotate symmetrically, creating pure stretching motion along the central stagnation point.

02 / CONTINUOUS SPECTRUM

Intermediate & Shear Flows (α=0)

By continuously tuning roll speed ratios, a full spectrum ($-1 < \alpha < 1$) of intermediate regimes is generated, passing through shear flow.

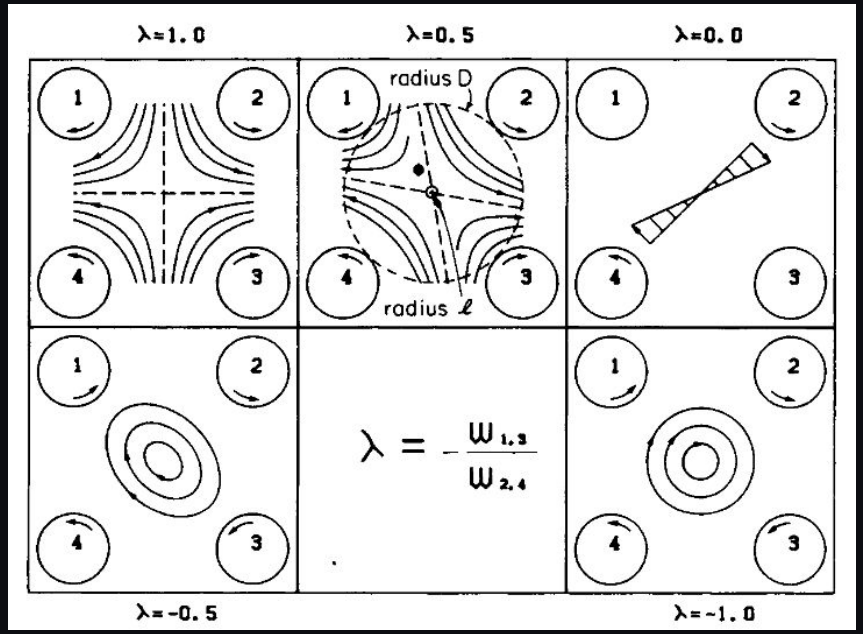
03 / FLOW REGIME

Rotational Flow (α=-1)

Achieved by driving rolls to induce co-rotational vorticity, placing fluid elements in pure rigid-body rotation.

Reference: Fuller et al. (1981) — Flow birefringence of concentrated polymer solutions in two-dimensional flows

4-Roll Mill Flow Fields



01 / KINEMATIC RELATION

$$\omega 1 = \omega 3 = \Omega, \quad \omega 2 = \omega 4 = -\alpha\Omega \quad (-1 \le \alpha \le 1)$$

$\Omega > 0$ (strain rate/scale of deformation), α flow regime parameter)

α	Velocity $(\omega_1, \omega_2, \omega_3, \omega_4)$	Flow Regime
1	$(\Omega, -\Omega, \Omega, -\Omega)$	Planar extensional
0	$(\Omega, 0, \Omega, 0)$	Simple shear
-1	$(\Omega, \Omega, \Omega, \Omega)$	Pure rotational

Reference: Fuller et al. (1981) — Flow birefringence of concentrated polymer solutions in two-dimensional flows

From Traditional CFD to Inference: Limits in Inverse Problems

01

THE INVERSE PROBLEM

Parameter Estimation

Aims to estimate unknown constitutive parameters (relaxation time λ, solvent/polymer viscosity η_s, η_p).

Inferred directly from experimental field measurements (e.g., PIV velocity fields).

02

FEM / CFD BOTTLENECKS

Classical Limitations

- **Iterative Process:** Depends on complex trial-and-error or adjoint sensitivity analysis.
- **Prohibitive Cost:** Requires full nonlinear meshed solves at every optimization step.
- **Numerical Instability:** Divergence at high Weissenberg numbers (Wi) breaks optimizers.

03

NEW PARADIGM

Physics-Informed NNs

Replaces pure mesh-based solvers with continuous neural representations (PINNs).

Natively incorporates experimental data and differential physical constraints into the loss function for seamless parameter inference.

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Literature Gap

01

CURRENT STATE

Simple Benchmarks

Recent developments in PINNs for viscoelastic (VE) fluids have been demonstrated only on simple benchmark problems (such as ViscoelasticNet).

02

THE GAP

Complex Geometries

There is a need to bridge this gap by extending models to handle complex geometries with realistic, relevant flow features.

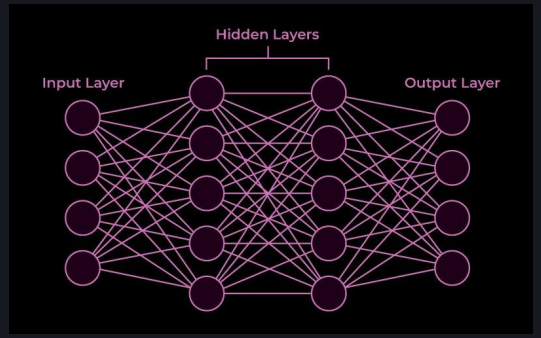
03

OBJECTIVE

Generalization

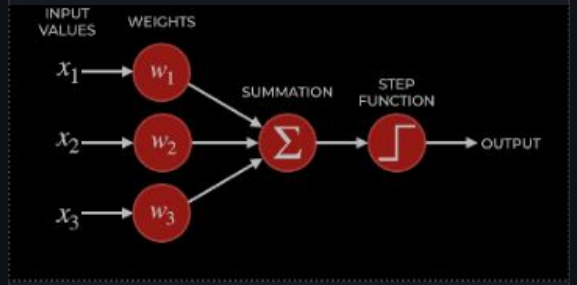
Developing robust neural network architectures capable of generalizing across diverse physical cases and complex fluid dynamics scenarios.

Fundamentals of Deep Learning



01 / ARCHITECTURE

Fully Connected



02 / PERCEPTRON

Activation Function

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i}^{N}(y_i - \hat{y}_i)^2$$

$$\theta_{t+1} = \theta_t - \eta\nabla_\theta\mathcal{L}(\theta)$$

03 / OPTIMIZATION

Loss & Gradient Descent

Reference: Goodfellow, I., Bengio, Y., & Courville, A. (2016) — *Deep Learning*, MIT Press.

Why Neural Networks?

01 CAPABILITIES

Cross-Domain Success

Neural networks have demonstrated exceptional problem-solving capabilities across various scientific, engineering, and data-driven domains.

02 FUNCTION APPROXIMATION

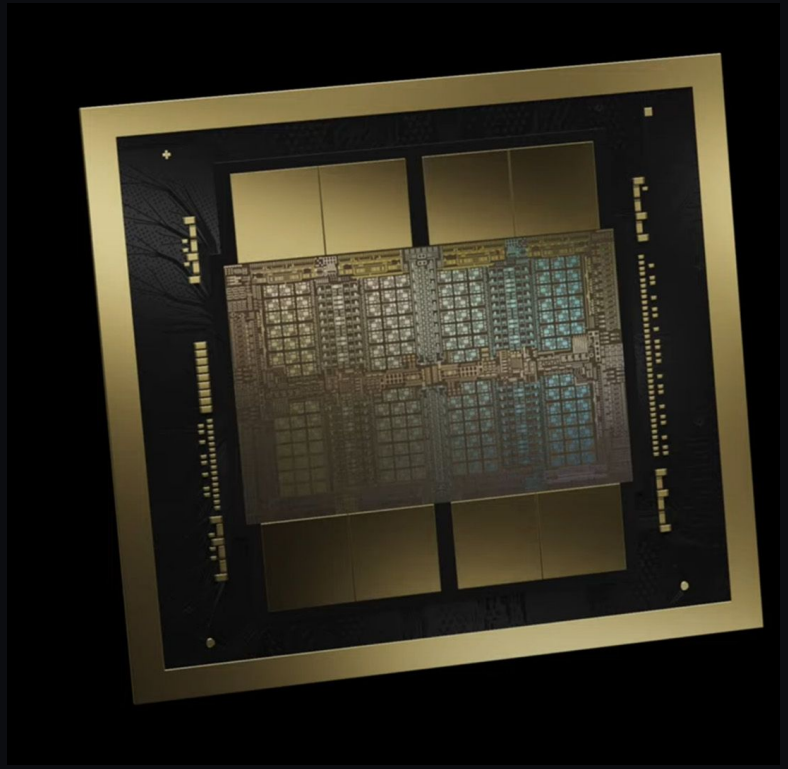
High-Dimensional Modeling

Acts as universal approximators capable of handling non-linear differential equations and high-dimensional continuous systems without grid constraints.

03 HARDWARE SYNERGY

Computational Progress

Modern computational hardware (GPUs, TPUs) is heavily optimized for parallel tensor operations, aligning directly with deep learning architectures.



NVIDIA Blackwell B200 GPU Die

Limits of Neural Networks

01

DATA DEPENDENCY

Limited Data Availability

Standard neural networks rely heavily on large volumes of labeled data, performing poorly or overfitting in data-scarce domains.

02

PHYSICAL CONSISTENCY

No Physical Grounding

Purely data-driven models lack embedded physical laws, often predicting solutions that violate fundamental physical constraints.

03

TRANSPARENCY

Interpretability

Complex architectures lack interpretability, making it difficult to analyze failure modes or trust outputs in critical applications.

Physics-Informed Neural Networks (PINNs)

Hybrid Paradigm

Bridges the structural gap between classical numerical PDE solvers and flexible modern deep learning architectures.

Loss Regularization

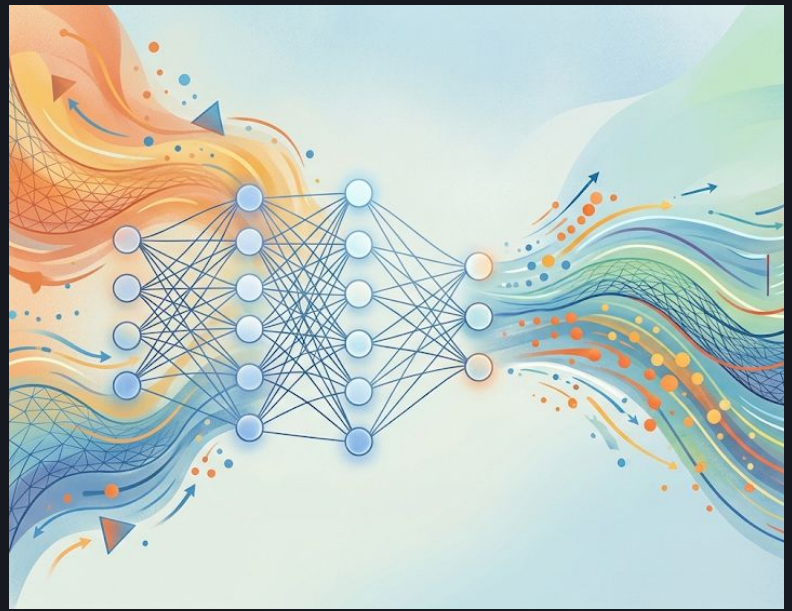
Embeds partial differential equations (PDEs) directly into the neural network's loss function to serve as an exact regularization mechanism.

Data Efficiency

Drastically reduces the volume of large labeled datasets required for training while enforcing continuous physical consistency.

Model Explainability

Provides physical interpretability and transparency by grounding neural representations in governing physical laws.



Reference: Raissi et al. (2019) — Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations

PINN Framework – Physical Information

$$L_{tot} = L_{data} + L_{physics}$$

DATA-DRIVEN COMPONENT

Loss of Data Mismatch

Penalizes difference against sparse experimental observations.

- **Supervised Learning:** Evaluates Mean Squared Error (MSE) at discrete measurement points.
- **Empirical Grounding:** Ensures model predictions conform strictly to observed sensor data & initial/boundary conditions.

PHYSICS-INFORMED COMPONENT

Loss of Physics Residuals

Enforces governing PDE laws at unlabelled collocation points.

- **Exact Differentiation:** Computed across domain collocation points via Automatic Differentiation (AD).
- **PDE Regularization:** Penalizes violations of differential conservation laws (e.g. Navier-Stokes).
- **Inverse problem:** Infer the physical parameters that best suit experimental data

PINN Framework – Automatic Differentiation

THE ANALOGY

Backpropagation vs. Coordinate AD

Just as standard gradient descent uses automatic differentiation to compute parameter gradients, it's possible to use computational graph to compute derivatives with respect to spatial inputs.

EXACT DIFFERENTIATION

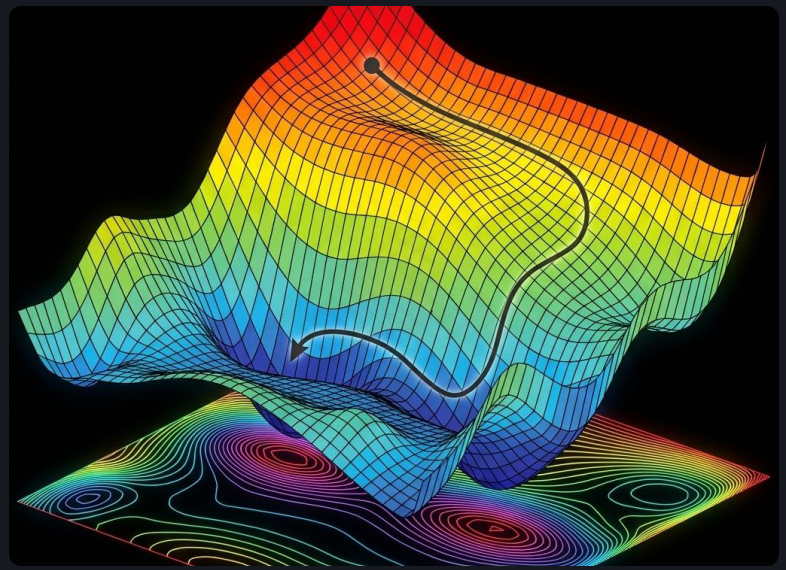
Pointwise Differential Operators

Evaluates exact PDE residuals $(\partial u/\partial t, \nabla^2 u)$ at continuous coordinates without truncation errors, avoiding any finite-difference stencil.

FUNDAMENTAL PROPERTY

Meshless Formulation

Completely decouples differential evaluation from geometric topology: collocation points require no cell connectivity, adjacency lists, or domain re-meshing.



PARAMETER SPACE OPTIMIZATION

Loss surface $\mathcal{L}(W, b)$ optimized via gradient descent, driving the physics-based PDE residual to zero across all collocation points.

Reference: Baydin et al. (2018) — Automatic differentiation in machine learning: a survey

Navier-Stokes Case Study – Flow Past a Cylinder

Problem Setup, Physical Assumptions & Flow Regime

FLUID PROPERTIES

Physical Assumptions

- **Newtonian & Incompressible:** Constant viscosity, $\nabla \cdot u = 0$
- **Governing Model:** 2D Navier-Stokes equations

GEOMETRY & DOMAIN

2D Flow Setup

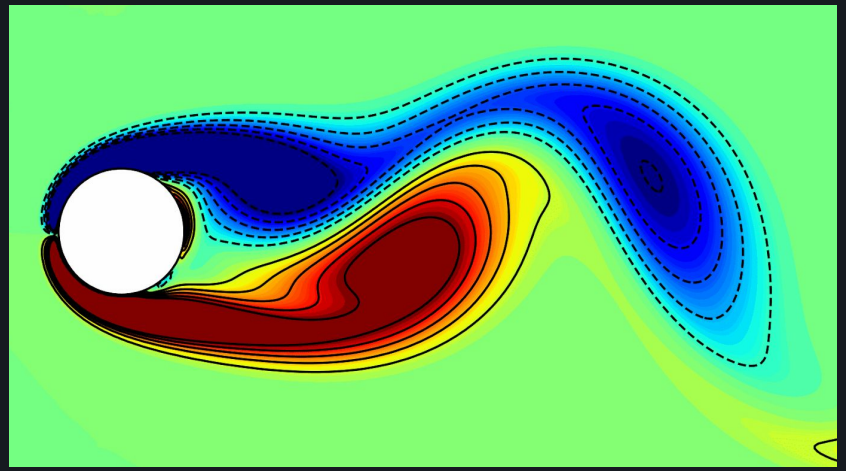
- **Obstacle:** Circular rigid cylinder with downstream wake
- **Sparse Sampling:** N = 5,000 points (1% space-time)

FLOW REGIME

Reynolds Number (Re = 100)

- **Vortex Shedding:** Unsteady Kármán vortex street
- **Inverse Identification:** Parameter recovery (λ_1, λ_2)

FLOW VISUALIZATION



Reference: Raissi et al. (2019) — Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations

Navier-Stokes Case Study – Physical vs. PINN Formulation

Inverse Parameter Identification & Physical Model

REAL PHYSICAL PROBLEM

Real Navier-Stokes Equations

Momentum Balance:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{Re}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)$$

Continuity Equation:

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

PINN FORMULATION

Learned Parameterized Equations

The network learns latent fields (u, v, p) and concurrently estimates unknown parameters λ_1 and λ_2.

Residual Loss Formulation:

$$f := u_t + \lambda_1(uu_x + vu_y) + p_x - \lambda_2(u_{xx} + u_{yy}) = 0$$

$$g := v_t + \lambda_1(uv_x + vv_y) + p_y - \lambda_2(v_{xx} + v_{yy}) = 0$$

Exact Continuity Satisfaction (ψ):

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x} \implies u_x + v_y \equiv 0$$

Inverse Parameters Sought by Model:

- λ_1 **(Convective Term):** Models inertial advective transport.
- λ_2 **(Diffusive Viscosity):** Corresponds to $1/Re$ in the Newtonian case.

Navier-Stokes Case Study – Modeling and Training Parameters

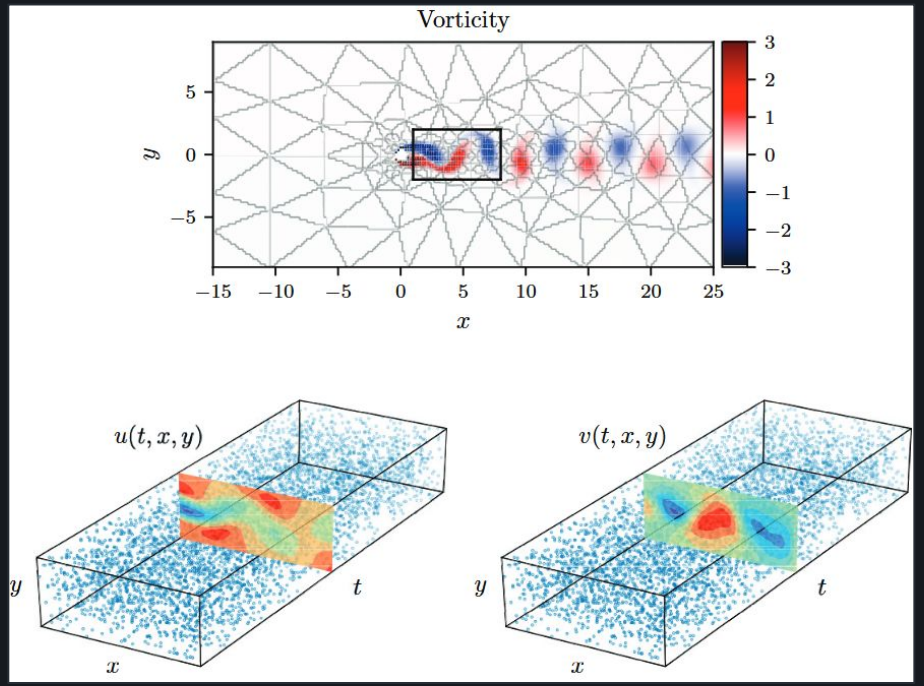
HYPERPARAMETERS & SETUP

Model Architecture

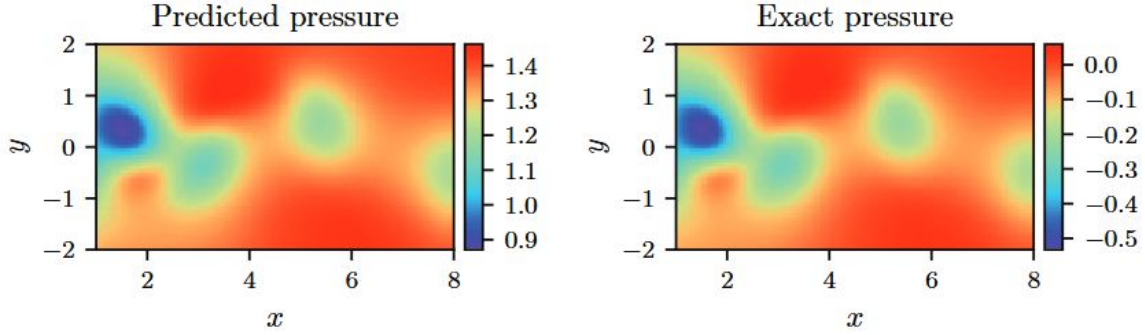
- **Activation**
  tanh
- **Layers & Neurons**
  1 network with 9 FC layers × 20 neurons, 2 outputs
- **Optimization**
  Adam(200k) + L-BFGS (50k)
- **Learning Rate**
  $\eta_o = 10^{-3}$ fixed
- **Sampling Points**
  N = 5,000 (1% of total data)

REPRESENTATIVE FLOW FIELDS

Velocity (u, v) data points



Navier-Stokes Results – Accuracy & Pressure Field



QUANTITATIVE VALIDATION

Robust Parameter Identification

Inverse estimation of physical coefficients λ_1 and λ_2:

- Clean Data: λ_1 error 0.08%, λ_2 error 4.67%
- 1% Gaussian Noise: λ_1 error 0.17%, λ_2 error 5.70%

High stability under experimental measurement noise.

KEY BREAKTHROUGH

"Hidden" Pressure Field Reconstruction (p)

- **Zero Pressure Training Data:** No pressure measurements were supplied during training.
- **Physics-Driven Inference:** The complete scalar pressure field p(t, x, y) is inferred strictly via pressure gradient constraints in momentum equations.

Matches pressure reference up to an additive constant.

ViscoelasticNet

CORE DOMAIN

Viscoelastic Fluid Modeling

KEY CHARACTERISTICS

- Non-Newtonian Flow
- Non-Linear Stresses
- High Weissenberg Instabilities

INVERSE FORMULATION

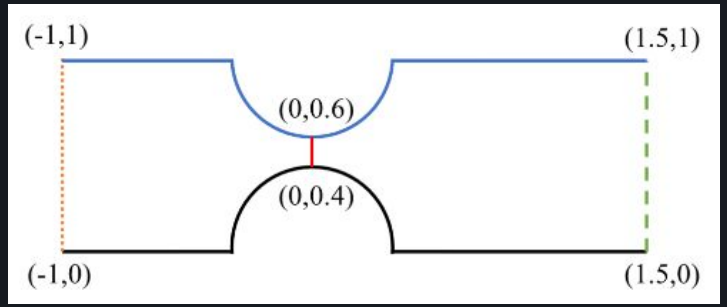
Parameters & VE Model Selection

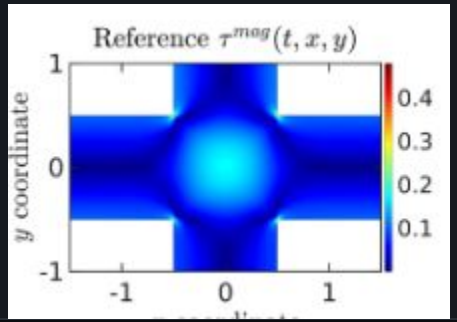
CAPABILITIES

- **Parameter Estimation:** Learns relaxation time (λ) and viscosity parameters ( ) directly from data.
- **Constitutive Discovery:** Classifies Oldroyd-B, Giesekus, or PTT model dynamics.
- **Data-Driven:** Reconstructs stress fields from sparse velocity observations.

BENCHMARK CASE STUDIES

Stenosis & Cross-Slot Flows





Reference: Thakur et al. (2024) - ViscoelasticNet: A physics informed neural network framework for stress discovery and model selection

ViscoelasticNet: Constitutive model(s)

UNIFIED CONSTITUTIVE FORMULATION

$$\left(1 + \frac{\epsilon\lambda}{\eta_p}\mathrm{tr}(\tau)\right)\tau + \lambda\overset{\nabla}{\tau} + \alpha\frac{\lambda}{\eta_p}(\tau\cdot\tau) = \eta_p\left(\nabla u + \nabla u^T\right)$$

$$\overset{\nabla}{\tau} = \frac{D\tau}{Dt} - (\nabla u)^T\cdot\tau - \tau\cdot(\nabla u)$$

$$\frac{D\tau}{Dt} = \frac{\partial\tau}{\partial t} + u\cdot\nabla\tau$$

MODEL SELECTION CRITERIA

ϵ = 0, α = 0

Oldroyd-B Model

Standard quasi-linear fluid model with constant viscosity and relaxation time.

ϵ = 0, α ≠ 0

Giesekus Model

Includes non-linear anisotropic drag effects via mobility parameter (**α**).

ϵ ≠ 0, α = 0

Linear PTT Model

Phan-Thien–Tanner model incorporating polymer extensibility parameter (**ϵ**).

ϵ ≠ 0, α ≠ 0

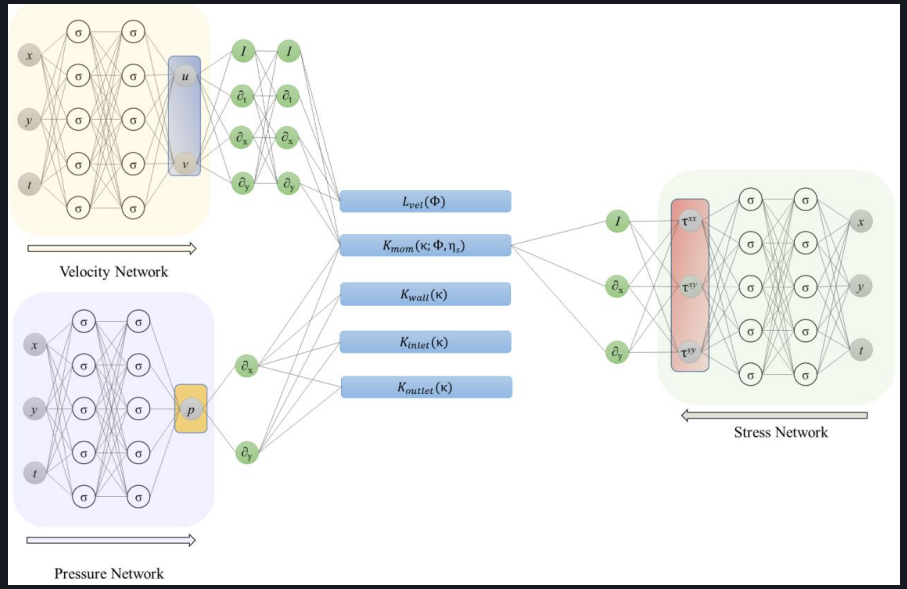
None of the Three

Model rejected for canonical subset (e.g., validated via *XPomPom* test cases).

Multi-Network Architecture & Decoupled Optimization

ARCHITECTURE & PIPELINE

3 Dedicated Neural Networks



DECOUPLED OPTIMIZATION

Two-Stage Sequential Training

STAGE 1: Stress & Rheology

Joint optimization of velocity, stress tensor, and material rheological parameters **(λ, ε, α, η_p)** via $\mathscr{L}$_stress loss.

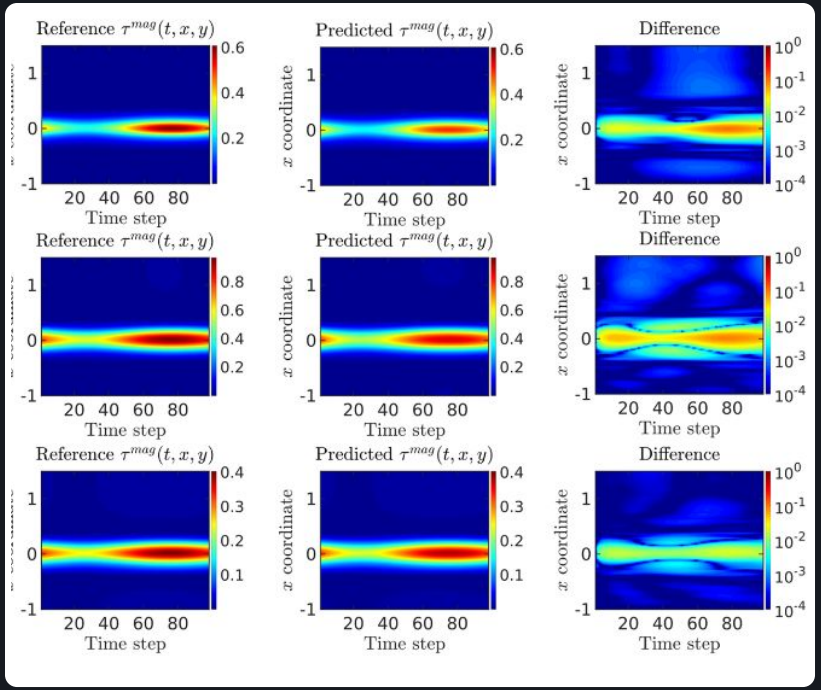
STAGE 2: Pressure & Viscosity

Freeze Stress Network weights; train Pressure Network and estimate solvent viscosity η_o by minimizing momentum residual.

Reference: Thakur et al. (2024) - ViscoelasticNet: A physics informed neural network framework for stress discovery and model selection

ViscoelasticNet: 2D Stenosis Results

STRESS FIELD COMPARISON



PARAMETER IDENTIFICATION UNDER NOISE

	α	ϵ	λ	η_p	η_s
Reference value	0.00	0.0	0.05	0.008	0.01
0% noise	0.00	0.0	0.0517	0.0081	0.0098
1% noise	0.00	0.0	0.0517	0.0081	0.0115
5% noise	0.00	0.0	0.0517	0.0081	0.0115
10% noise	0.00	0.0	0.0517	0.0081	0.0112

	α	ϵ	λ	η_p	η_s
Reference value	0.00	0.1	0.15	0.015	0.01
0% noise	0.00	0.106	0.161	0.0157	0.0097
1% noise	0.00	0.108	0.162	0.0157	0.0123
5% noise	0.00	0.108	0.162	0.0157	0.0123
10% noise	0.00	0.108	0.161	0.0157	0.0127

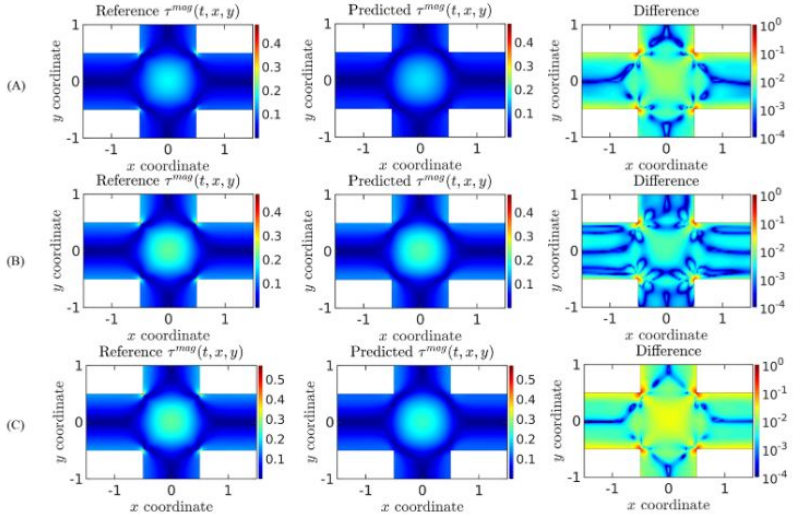
FIELD ACCURACY & RELATIVE ERROR FORMULATION

	u	v	p
0% Noise	1.3×10^{-3}	2.0×10^{-3}	1.95×10^{-1}
1% Noise	1.4×10^{-3}	2.1×10^{-3}	1.82×10^{-1}
5% Noise	1.4×10^{-3}	2.2×10^{-3}	1.83×10^{-1}
10% Noise	1.7×10^{-3}	2.5×10^{-3}	1.77×10^{-1}

$$\mathcal{L}(a_{reference}, a_{prediction}) = \sqrt{\frac{(a_{reference} - a_{prediction})^2}{(a_{reference} - \bar{a}_{reference})^2}},$$

ViscoelasticNet: Cross-Slot Flow Results

STRESS FIELD COMPARISON



PARAMETER IDENTIFICATION & PREDICTED VALUES

	α	ϵ	λ	η_p	η_s
Giesekus Reference value	0.05	0.0	0.004	0.003	0.01
Giesekus Predicted value	0.056	0.0	0.00386	0.0273	0.011
Linear PTT Reference value	0.0	0.02	0.008	0.025	0.01
Linear PTT Predicted value	0.0	0.0183	0.0085	0.0245	0.0099
Linear PTT Reference value #2	0.0	0.02	0.008	0.025	0.01
Linear PTT Predicted value #2	0.0	0.0228	0.00836	0.0239	0.0102
Oldroyd-B Reference value	0.0	0.0	0.005	0.01	0.01
Oldroyd-B Predicted value	0.0	0.0	0.0046	0.0188	0.011
Oldroyd-B Reference value #2	0.0	0.0	0.015	0.01	0.02
Oldroyd-B Predicted value #2	0.0	0.0	0.0135	0.0193	0.0236
Oldroyd-B Reference value #3	0.0	0.0	0.01	0.025	0.02
Oldroyd-B Predicted value #3	0.0	0.0	0.0093	0.033	0.0171
XPomPom Predicted value	37.29	0.00423	0.29	0.145	0.011

Overview of our work

01 PARAMETER EXTRACTION

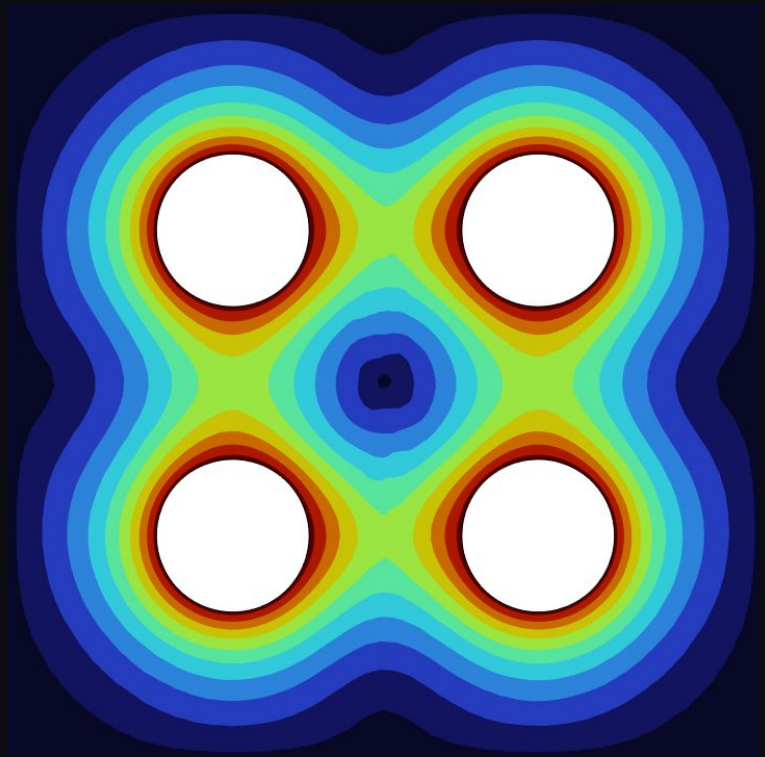
Physical Parameters

PINN applied to a four-roll mill geometry to extract key physical parameters from experimental fluid dynamics data.

02 MODEL SELECTION

Constitutive Modeling

Identification and evaluation of the constitutive model that best fits and describes experimental observations.



Differences from ViscoelasticNET

01

REGIME TYPE

Steady-State

Formulated exclusively for stationary conditions ⟶ **∂/∂t = 0**.

Neglects transient fluid dynamics to isolate steady flow behavior and optimize computational focus.

02

SCOPE & STAGE

Phase 1 Focus

Restricted to initial baseline configurations ⟶ **Phase 1**.

Constitutive Models:

Focuses exclusively on **Oldroyd-B** and **Giesekus** models.

03

OPERATIONAL BOUNDS

Extended Ranges

Significantly wider parameter space vs ViscoelasticNET benchmark:

λ: up to 1 s (vs 0.15 s, **10×**)

η_p: up to 0.98 Pa·s (vs ≤0.025 Pa·s, **40×**)

α: up to 0.50 (vs 0.20)

ViscoelasticNet: Results

VALIDATION GEOMETRIES

Benchmark Flow Cases

1. 2D Stenosis Flow

Hemodynamic pulsatile sinusoidal flow validation:

- **< 5% relative error** on stress fields.
- Accurate classification of Oldroyd-B, Linear PTT, and Giesekus models.

2. Cross-Slot Flow

Strong extensional flow benchmark:

- Correct identification of non-linear rheological parameters.
- Capturing stress singularities at sharp corners requires domain decomposition.

DATA ROBUSTNESS

Noise & Sparsity Tolerance

Gaussian Noise Resistance

- Tested with up to **10% Gaussian noise** on velocity fields.
- Up to **25% noise** on boundary conditions.
- Learned rheological parameters remain robustly stable.

Sparse Data Threshold

- Maintains high accuracy down to **5×10^4 total points**.
- Fully compatible with resolutions from real experimental measurements.

PRACTICAL IMPACT

Experimental PIV Integration??

The PIV Limitation

Laboratory Particle Image Velocimetry (PIV) only provides kinematic velocity fields; internal stresses and pressure are inaccessible directly.

Full-Field Virtual Rheometer

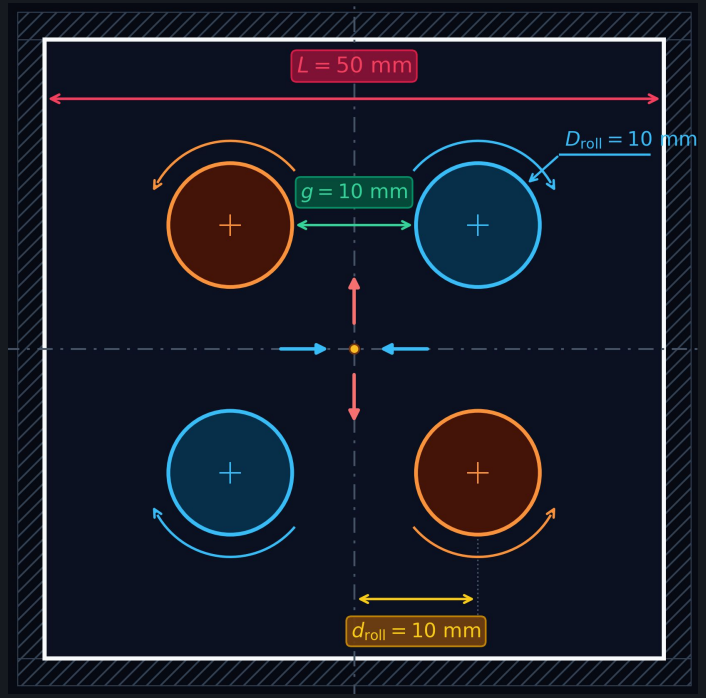
ViscoelasticNet acts as a **virtual rheometer**:

- Uncovers full stress tensors and pressure fields directly from raw PIV data.
- Identifies underlying constitutive equations automatically.

CFD Modeling: Geometric Ratios & Benchmark Setup

GEOMETRIC DOMAIN

Characteristic Dimensions



GEOMETRIC RATIOS

CYLINDER RATIO (Droll / L)

0.2

POSITIONING RATIOS (droll / L)

0.2

Reference: D'Avino et al. (2017) – *Numerical simulations of dispersive mixing of viscoelastic suspensions in a four-roll mill*

Physical Assumptions

01

FLUID DENSITY

Incompressible

Constant density $\longrightarrow$ $\nabla \cdot \mathbf{u} = 0$.

Automatically satisfied through the formulation of the stream function (ψ).

02

FORCE FIELDS

No Gravity

Negligible gravity forces $\longrightarrow$ $\mathbf{g} = 0$.

Isolates purely hydrodynamic forces and eliminates mass/body force terms in momentum balance.

03

THERMODYNAMICS

Isothermal

Uniform temperature profile $\longrightarrow$ $\mathbf{T = const}$ $(\nabla T = 0)$.

Decouples the energy equation and eliminates thermal dissipation effects.

Recap: Governing Equations & Boundary Conditions

01 CONTINUITY EQUATION

Incompressible Flow Regime

$$\nabla \cdot \boldsymbol{u} = 0$$

02 MOMENTUM BALANCE

Quasi-Stokes Steady-State

$$\rho\left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u}\right) = -\nabla p + \eta_s \nabla^2 \boldsymbol{u} + \nabla \cdot \boldsymbol{\tau}$$

03 RHEOLOGICAL CONSTITUTIVE EQUATION

Giesekus Model Formulation

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha\lambda}{\eta_p}(\boldsymbol{\tau} \cdot \boldsymbol{\tau}) = 2\eta_p \boldsymbol{D}$$

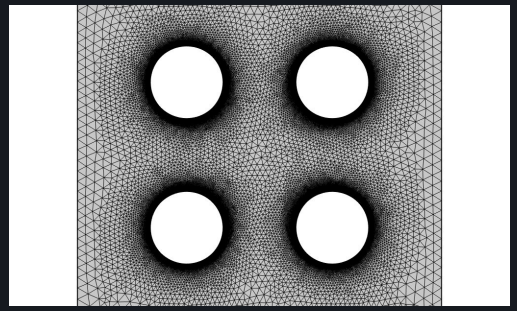
04 BOUNDARY CONDITIONS

Wall, Roll No-Slip & Pressure Reference

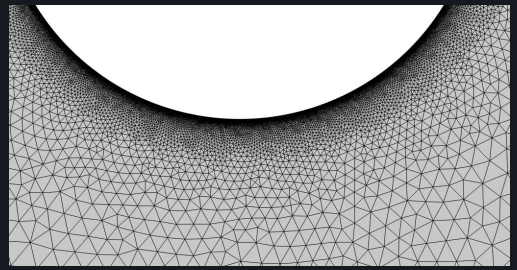
- Outer Walls (No-Slip): $u = 0$
- Rotating Rolls(No-Slip): $u = \pm\omega \cdot r$
- Pressure Point: $p = p_0$

CFD Meshes & Boundary Detail

125k

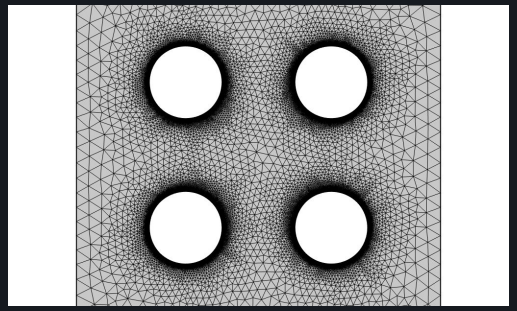


▼ DETAIL

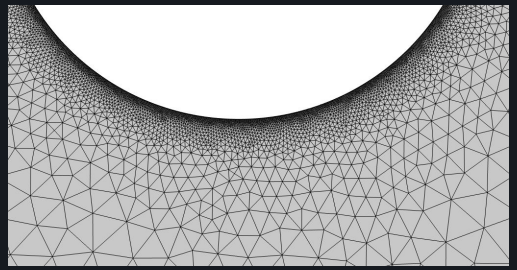


High boundary layer refinement

52k

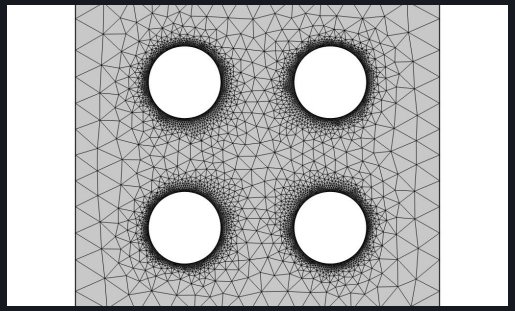


▼ DETAIL

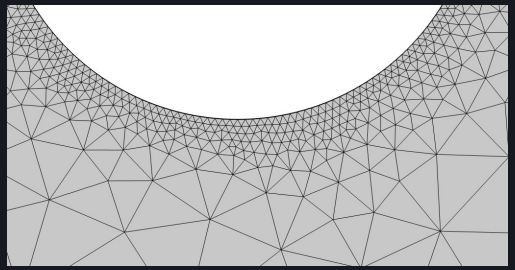


Balanced mesh density

5k



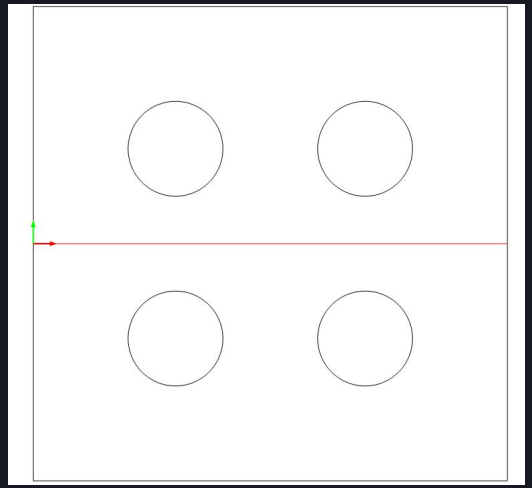
▼ DETAIL



Coarse discretization

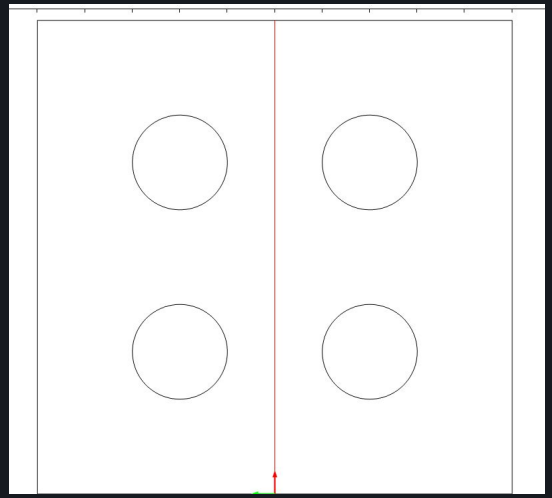
Relevant Cutlines

HORIZONTAL CUTLINE



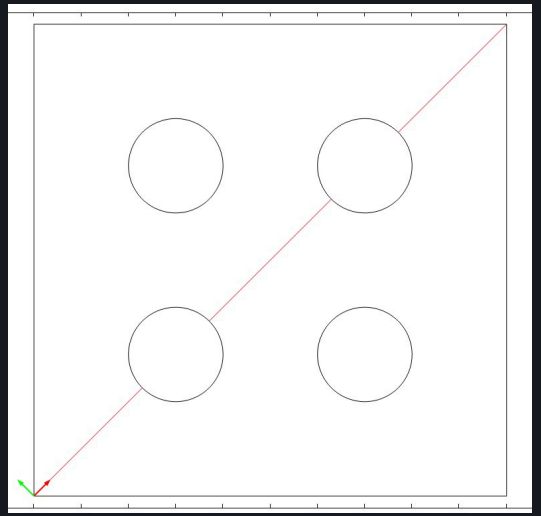
Horizontal cross-section passing through the domain center

VERTICAL CUTLINE



Vertical cross-section along the central symmetry axis

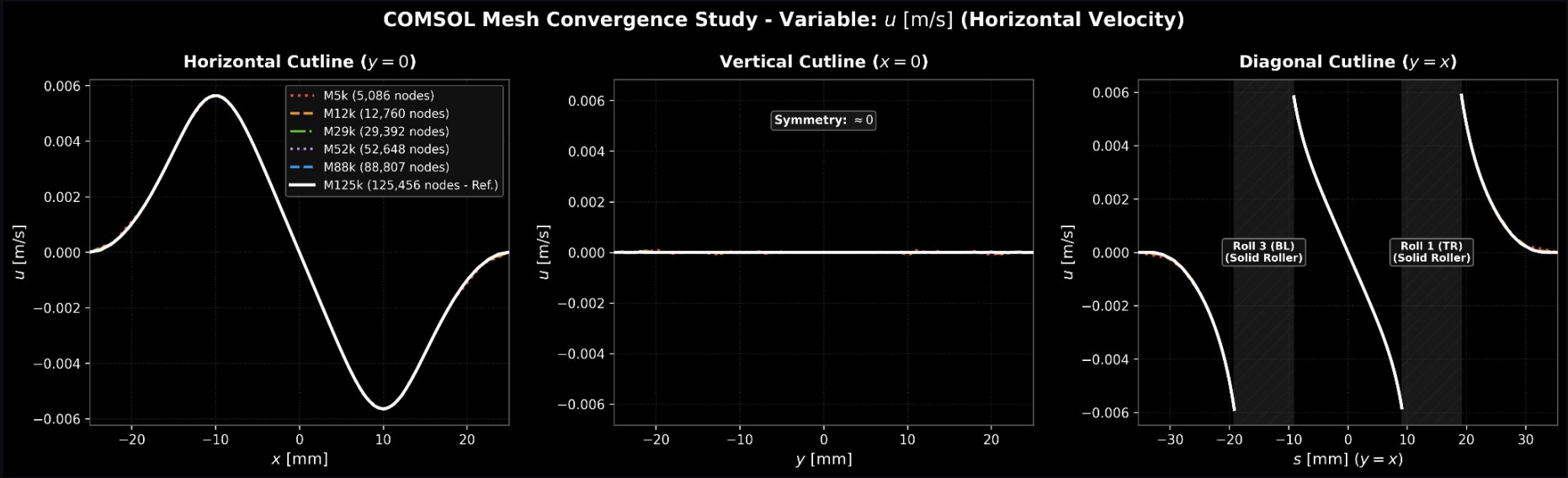
OBLIQUE CUTLINE



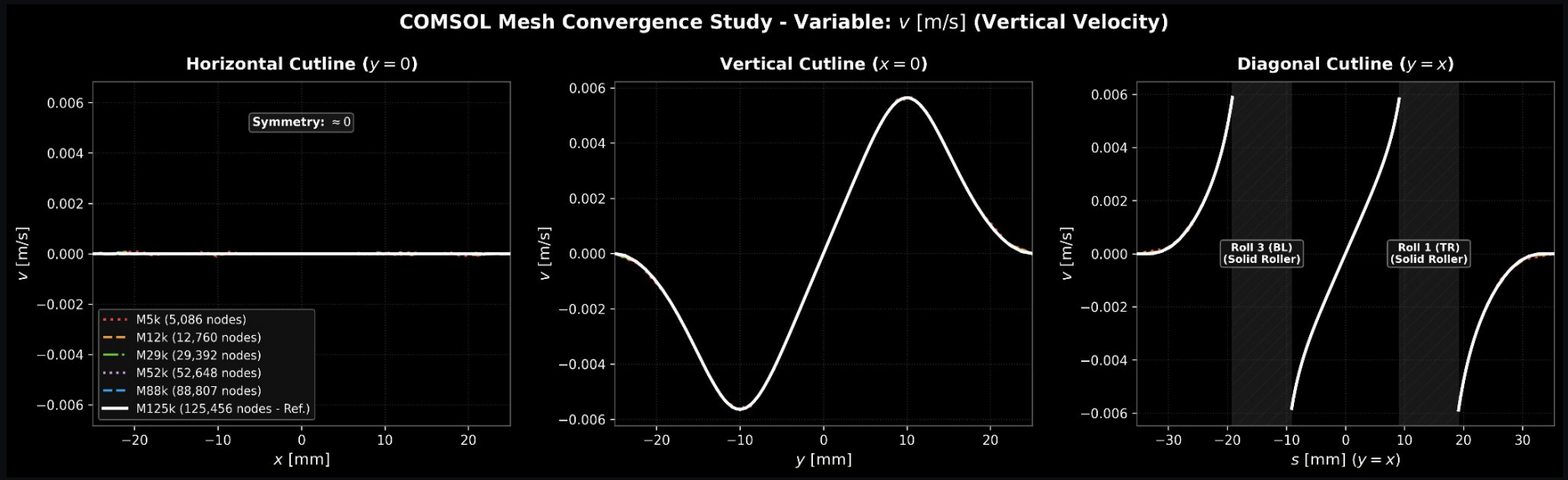
Diagonal cross-section traversing opposite domain corners

Horizontal Velocity (u)

SETUP: Mesh Convergence | MESH: All | PARAMS: λ =1 s, ηp = 0.9 Pa s

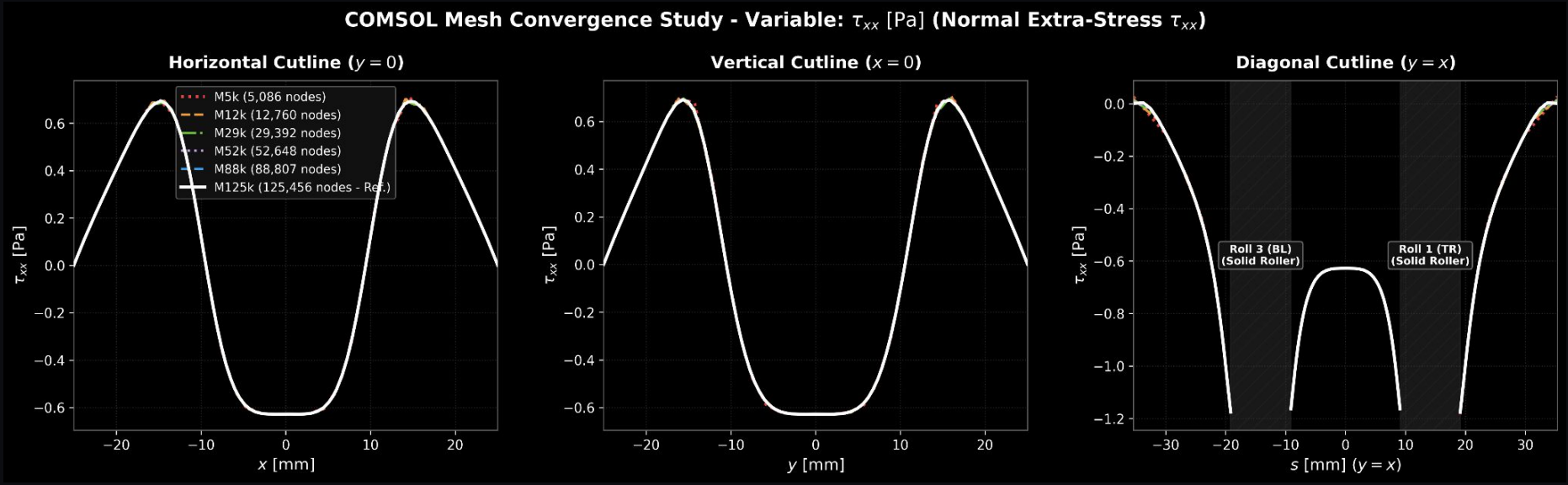


V


COMSOL Mesh Convergence Study - Variable: v [m/s] (Vertical Velocity)
Horizontal Cutline (y = 0)
Symmetry: ≈ 0
M5k (5,086 nodes)
M12k (12,760 nodes)
M29k (29,392 nodes)
M52k (52,648 nodes)
M88k (88,807 nodes)
M125k (125,456 nodes - Ref.)
v [m/s]
x [mm]
Vertical Cutline (x = 0)
v [m/s]
y [mm]
Diagonal Cutline (y = x)
Roll 3 (BL)
(Solid Roller)
Roll 1 (TR)
(Solid Roller)
v [m/s]
s [mm] (y = x)

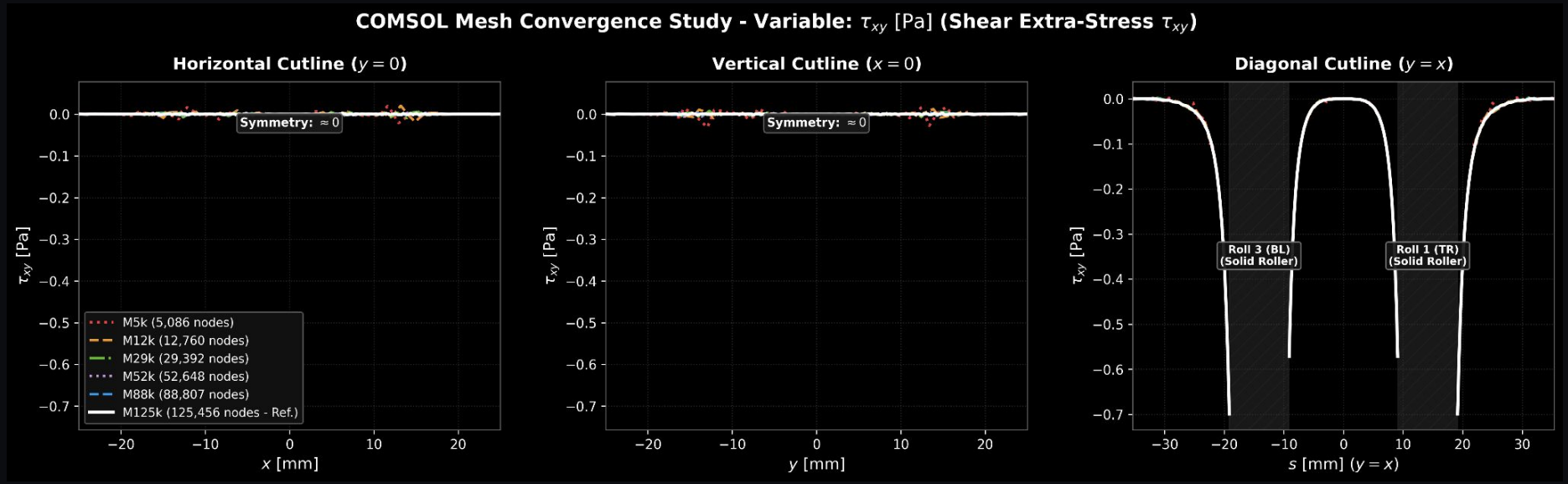
Normal extra stress (tauxx)

SETUP: Mesh Convergence | MESH: All | PARAMS: λ =1 s, ηp = 0.9 Pa s

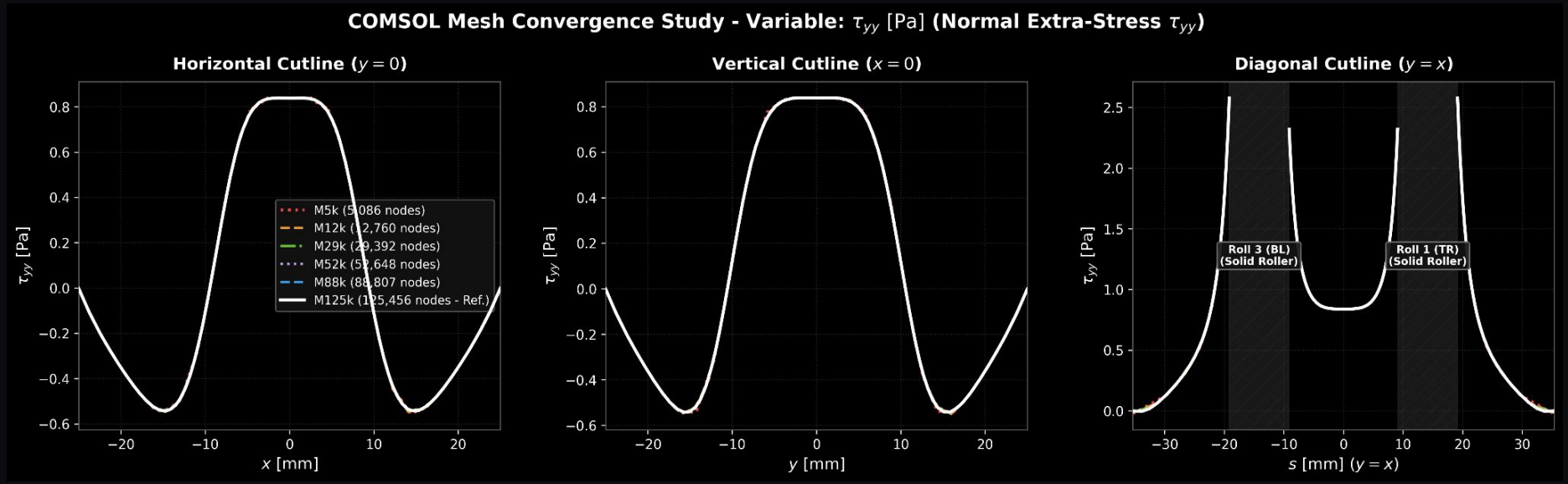


Shear stress (tauxy)

SETUP: Mesh Convergence | MESH: All | PARAMS: λ =1 s, ηp = 0.9 Pa s



tauyy


COMSOL Mesh Convergence Study - Variable: τ_{yy} [Pa] (Normal Extra-Stress τ_{yy})
Horizontal Cutline ($y = 0$)
Vertical Cutline ($x = 0$)
Diagonal Cutline ($y = x$)
τ_{yy} [Pa]
x [mm]
y [mm]
s [mm] ($y = x$)
M5k (5,086 nodes)
M12k (12,760 nodes)
M29k (29,392 nodes)
M52k (52,648 nodes)
M88k (88,807 nodes)
M125k (125,456 nodes - Ref.)
Roll 3 (BL) (Solid Roller)
Roll 1 (TR) (Solid Roller)

Physical parameters & Collocation Points

Physical Parameters

Roll Velocity (uroll): 8.333×10^{-3} m/s

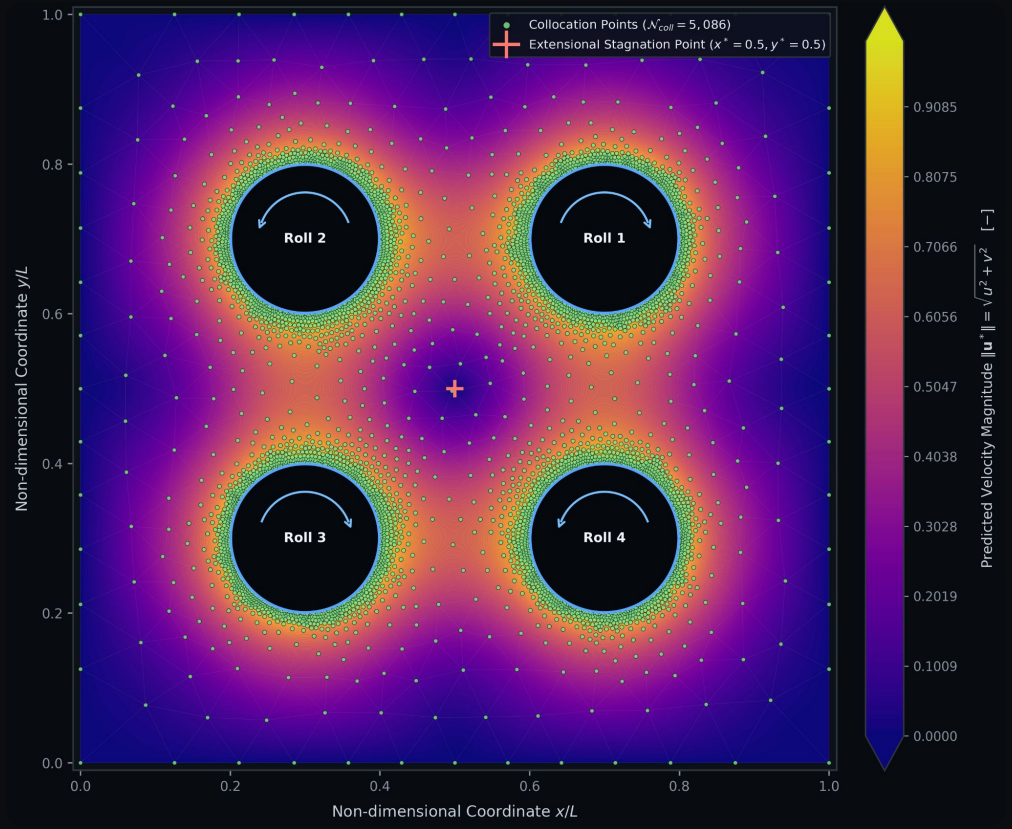
Density (ρ): 1000 kg/m³

Flow Characteristics:
Purely extensional flow field considering flow parameter

Reynolds Num. (Re): $0.0417 \ll 1$

Weissenberg Num. (Wi):

Deborah Num. (De):



Multi-Objective Loss Function & BC Formulation

`Ltotal = wdata · Ldata + wbc · Lbc + wconst · Lconst + wmom · Lmom`

1. Velocity Data Loss (Ldata)

$$\mathcal{L}_{\text{data}} = \frac{1}{N_u}\sum_{i=1}^{N_u}\|\mathbf{u}-\mathbf{u}_{\text{data}}\|^2$$

$$\text{con}\quad \mathbf{u} = \left(\frac{\partial\psi}{\partial y}, -\frac{\partial\psi}{\partial x}\right) \implies \nabla\cdot\mathbf{u}\equiv 0$$

2. Constitutive Eq. (Lconst)

$$\mathbf{r}_{\text{const}} = \boldsymbol{\tau} + \lambda\,\overset{\nabla}{\boldsymbol{\tau}} + f_{\text{nl}}(\boldsymbol{\tau}) - 2\eta_p\mathbf{D}$$

$$\mathcal{L}_{\text{const}} = \frac{1}{N_f}\sum_{i=1}^{N_f}\|\mathbf{r}_{\text{const}}\|^2$$

3. Momentum Balance (Lmom)

$$\mathbf{r}_{\text{mom}} = \rho\,(\mathbf{u}\cdot\nabla)\mathbf{u} + \nabla p - \eta_s\nabla^2\mathbf{u} - \nabla\cdot\boldsymbol{\tau}$$

$$\mathcal{L}_{\text{mom}} = \frac{1}{N_f}\sum_{i=1}^{N_f}\|\mathbf{r}_{\text{mom}}\|^2$$

4. Boundary Conditions (Lbc)

- **No-Slip Kinematics**
- **Stress Tensor Anchor**
- **Pressure Point Reference**

5. Staged Training Strategy

- **Phase 1 (Rheology & Kinematics):** wmom = 0, wconst > 0 → Trains stream function ψ, stress τ, and learns inverse parameters (λ, μp, α, ε).
- **Phase 2 (Hydrodynamics & Pressure):** wconst =0, wmom > 0, τ frozen → Solves pressure field p and solvent viscosity μs with ψ fine-tuned at reduced learning rate.

PINN Training Setup & Parameters [Direct / Forward Problem]

1. Network Architecture

Structure: 3 Decoupled Heads

Inputs/Outputs:
`(x,y) → ψ (1), p (1), τ (3: τxx, τxy, τyy)`

Layers: 8 hidden layers × 128 neurons

Activation: **SiLU** (smooth 2nd derivatives)

Weights: Xavier Normal (Zero-init p & τ last layer)

2. Staged Optimization Strategy

PHASE 1: Kinematics & Rheology
- Train ψ & τ; Freeze p (Wmom=0, Wcon=1.0)
- **Fixed Parameters:** λ, μp, α, ε (No inverse learning)

PHASE 2: Hydrodynamics & Pressure
- Freeze τ (precomputed ∇·τ); Train p & ψ
- Micro-unfreezing: ψ active at LR/10
- **Known solvent viscosity μs** (Wmom=1.0, Wcon=0)

4. Multi-Stage Precision & Solvers

Stage 1: Adam

Epochs: 40k (Scratch) / 20k (TL)

Base LR: 2.5 × 10-3

Scheduler: Cosine Annealing (down to ηmin = 2.5 × 10-6)

Precision: FP32

Stage 2: L-BFGS

Max Iter: 10k (Scratch) / 5k (TL)

Line Search: `strong_wolfe`

Precision: Full FP64 / Double conversion

5. Loss Weights & Infrastructure

Loss Weights Configuration:

```
Phase 1: Wdata=1.0 | Wbc=5.0 | Wcon=1.0 | Wmom=0.0 | Wroll_stress=1.0
Phase 2: Wdata=20.0 | Wbc=5.0 | Wmom=1.0 | Wcon=0.0
```

Reproducibility: Fixed Global Seed = 123

PINN Strategy: Direct vs. Inverse Problem Comparison

1. Network Architecture (Shared Baseline)

Structure: 3 Decoupled Heads | **Layers:** 8 hidden layers × 128 neurons | **Activation:** **SiLU**

Inputs / Outputs: `(x,y) → ψ (1), p (1), τ (3: τxx, τxy, τyy)` | **Epochs:** Adam 40k (20kTL)+ L-BFGS 10k (5kTL)

Base LR: 2.5 × 10-3 | **Scheduler:** Cosine Annealing (down to ηmin = 2.5 × 10-6)

Direct Problem (Forward)

PHASE 1: Kinematics & Rheology
- Train ψ & τ; Freeze p (Wmom=0, Wcon=1.0)
- **Fixed Parameters:** λ, μp, α, ε (No inverse learning)

PHASE 2: Hydrodynamics & Pressure
- Freeze τ (precomputed ∇·τ); Train p & ψ
- Micro-unfreezing: ψ active at LR/10
- **Known solvent viscosity μs** (Wmom=1.0, Wcon=0)

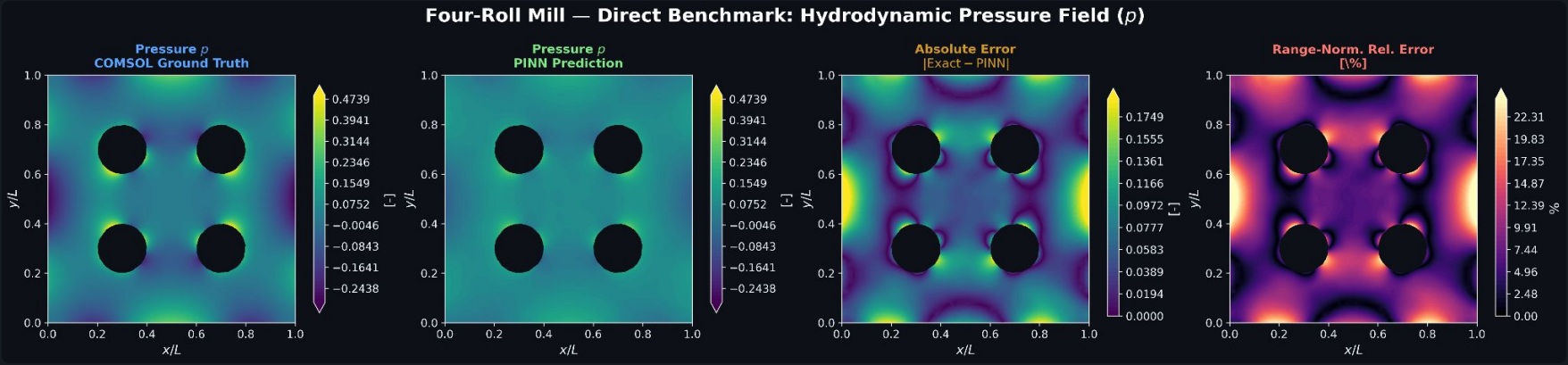
Inverse Problem (Parameter Learning)

PHASE 1: Kinematics & Rheology
- Train ψ & τ; Freeze p (Wmom=0, Wcon=1.0)
- **Active Parameters:** λ, μp, α, ε (Inferred online)

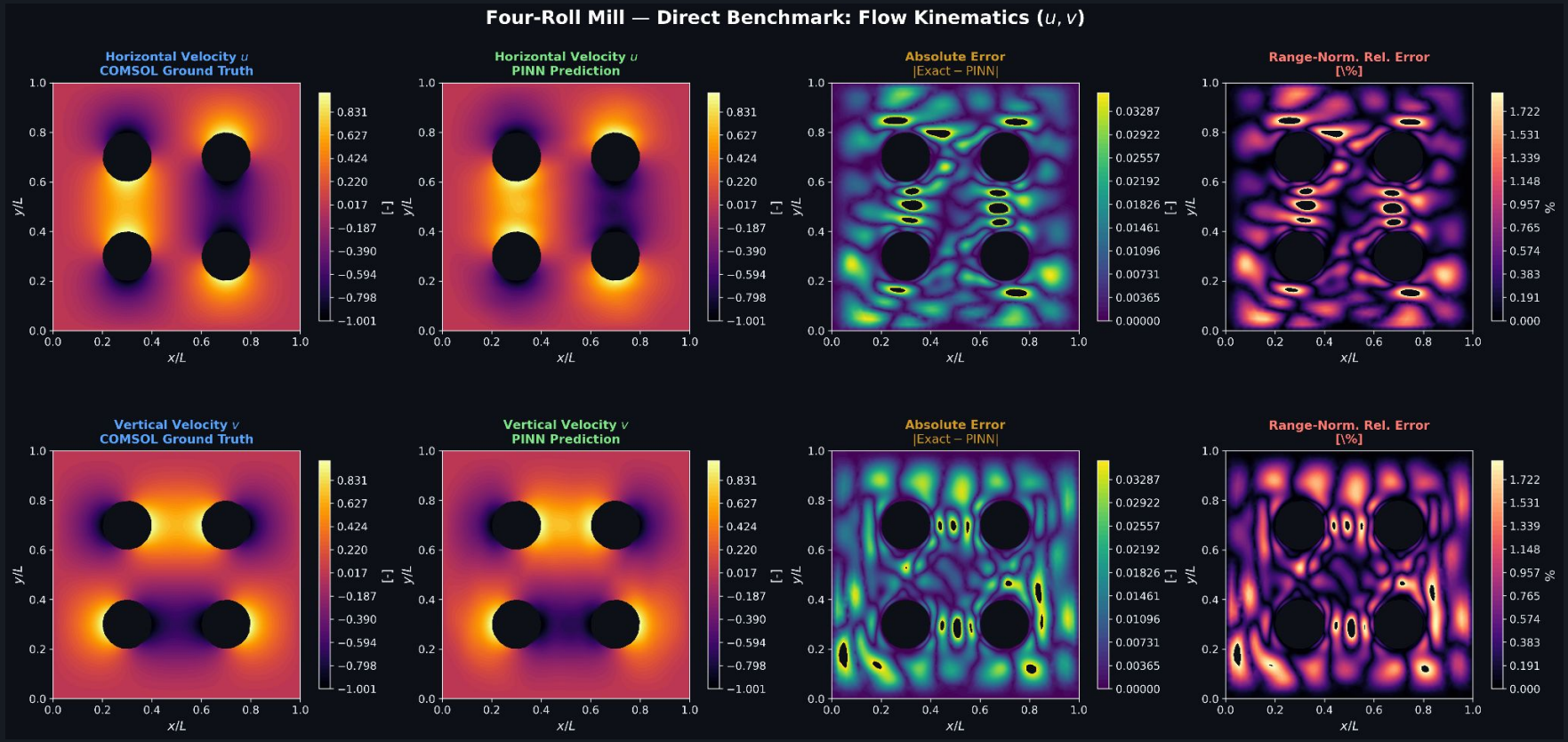
NO PHASE 2

Four-Roll Mill — Hydrodynamic Pressure (p)

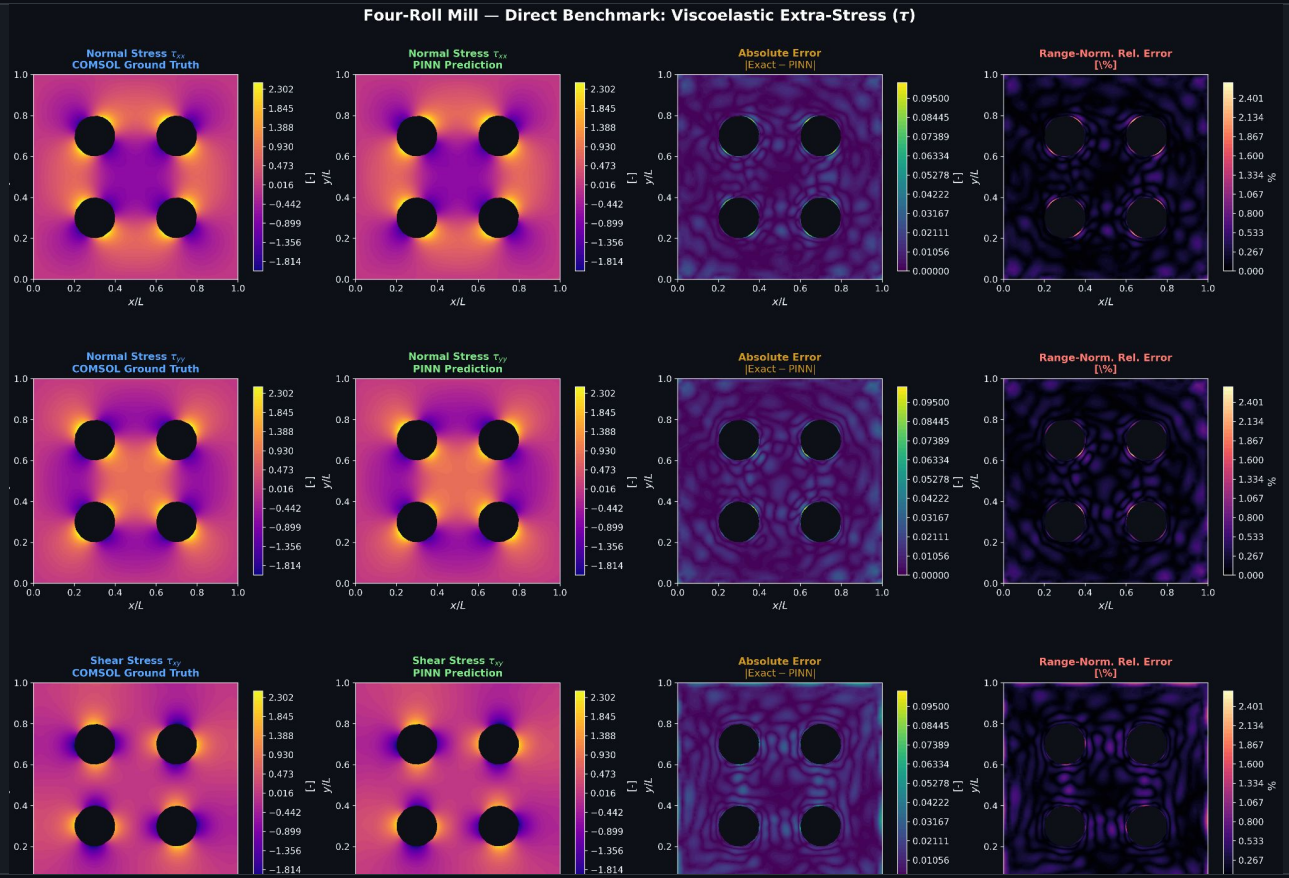
SETUP: Direct Problem | MESH: M5k | PARAMS: λ =0.05 s, ηp = 0.9 Pa s



Four-Roll Mill — Flow Kinematics (u, v)



Four-Roll Mill — Viscoelastic Extra-Stress (τ)



PINN Training Setup & Parameters [Inverse Problem]

1. Network Architecture

Structure: 3 Decoupled Heads

Inputs/Outputs:
`(x,y) → ψ (1), p (1), τ (3: τxx, τxy, τyy)`

Layers: 8 hidden layers × 128 neurons

Activation: **SiLU** (smooth 2nd derivatives)

Weights: Xavier Normal (Zero-init p & τ last layer)

2. Staged Optimization Strategy

PHASE 1: Kinematics & Rheology

- Train ψ & τ; Freeze p (Wmom=0, Wcon=1.0)
- **Active Parameters:** λ, μp, α, ε

NO PHASE 2

4. Multi-Stage Precision & Solvers

Stage 1: Adam

Epochs: 40k (Scratch) / 20k (TL)

Base LR: 2.5 × 10-3

Scheduler: Cosine Annealing (down to ηmin = 2.5 × 10-6)

Precision: FP32

Stage 2: L-BFGS

Max Iter: 10k (Scratch) / 5k (TL)

Line Search: `strong_wolfe`

Precision: Full FP64 / Double conversion

5. Loss Weights & Infrastructure

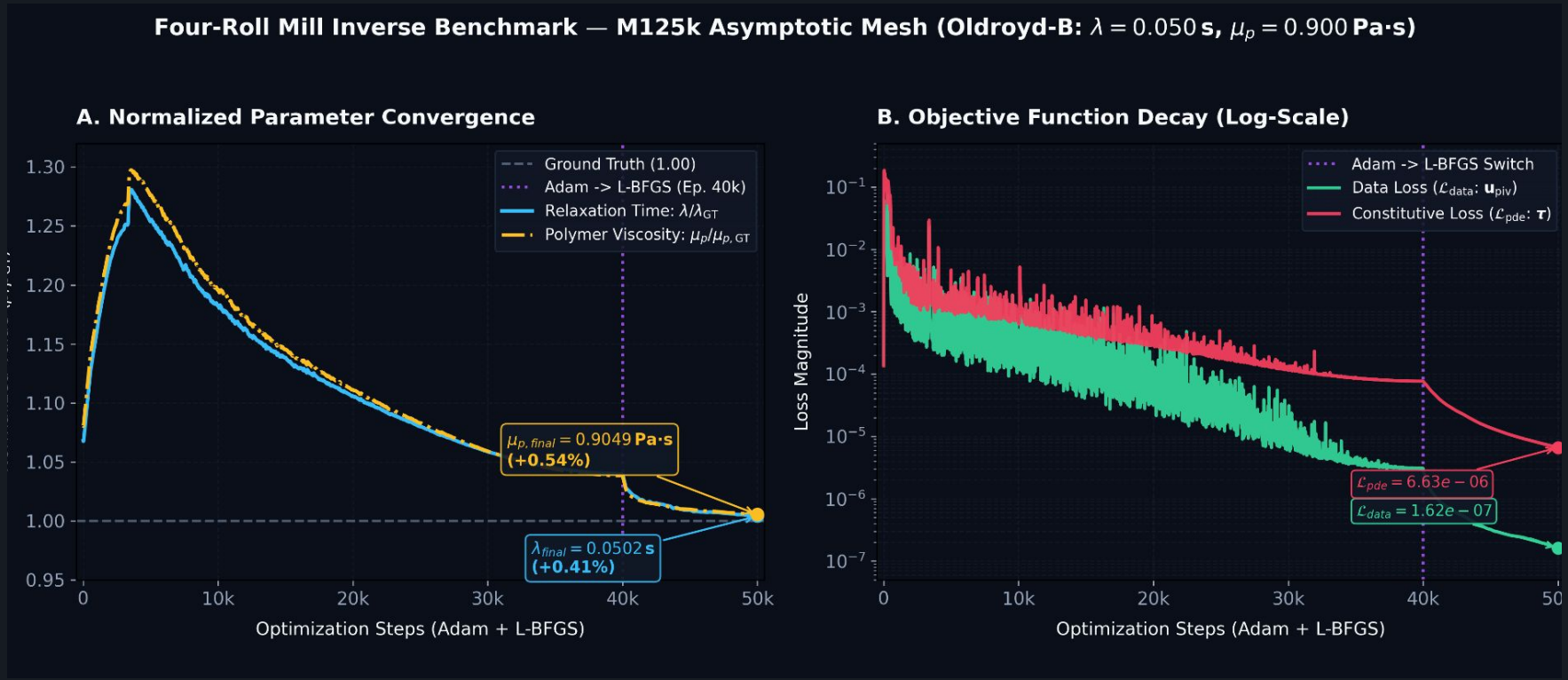
Loss Weights Configuration:

```
Phase 1: Wdata=1.0 | Wbc=5.0 | Wcon=1.0 | Wmom=0.0 | Wroll_stress=1.0
Phase 2: Wdata=20.0 | Wbc=5.0 | Wmom=1.0 | Wcon=0.0
```

Reproducibility: Fixed Global Seed = 123

Four-Roll Mill — Inverse Problem Results

SETUP: Inverse Parameter Estimation | MESH: M125k (Highest Resolution) | PARAMS: λ =0.05 s, ηp = 0.9 Pa s

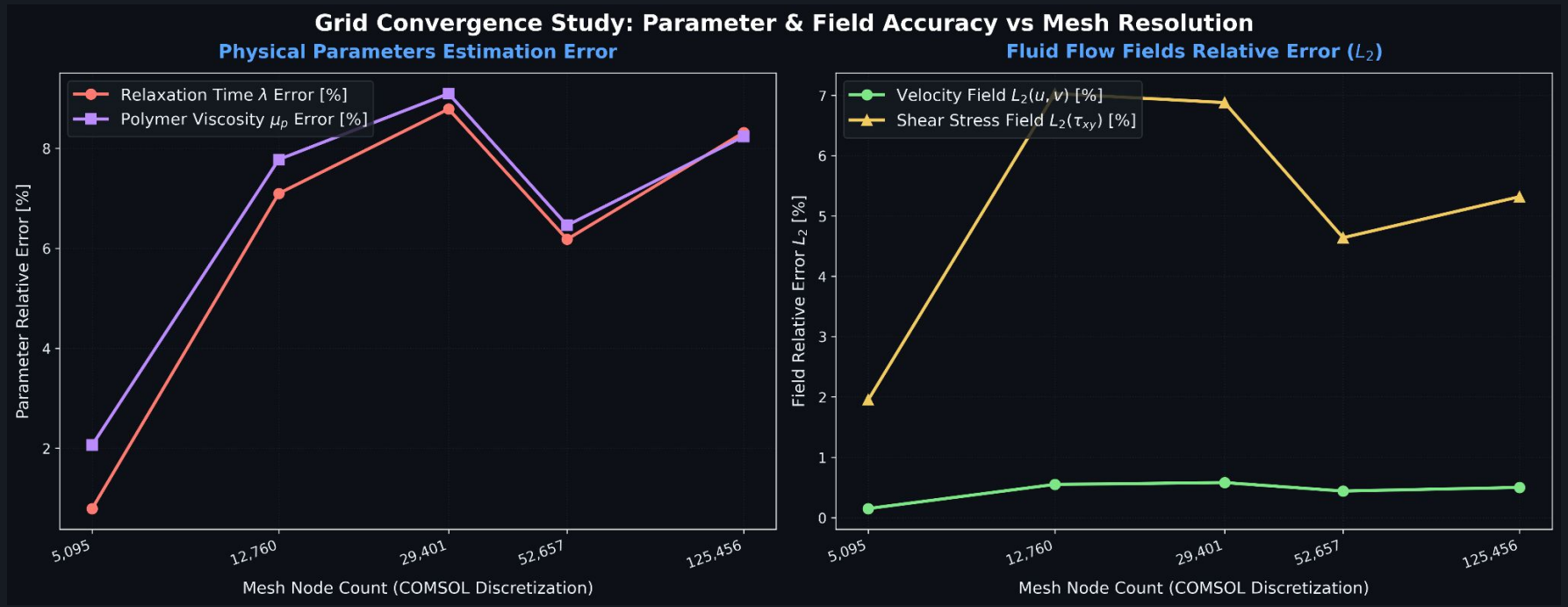


Inverse Problem: Key Parameter Discovery Benchmarks

Case #	Constitutive Model & Setup	Training Strategy & TF Source	Target Parameter	Ground Truth	PINN Prediction	Parameter Error	Velocity Error L_2
Case 1	**Canonical Oldroyd-B (Standard Low-$Wi = 0.08$)**	Ex-Novo (From Scratch) 40k Adam + 10k L-BFGS Donor: None	Relaxation time λ [s] Polymer viscosity μ_p [Pa·s]	0.0500 0.9000	**0.0502** **0.9049**	**+0.41%** **+0.54%**	**0.040%**
Case 2	**Oldroyd-B High Elasticity (High-$Wi = 1.17$, Boger Fluid)**	Transfer Learning (TF) 20k Adam + 5k L-BFGS (-50% budget) Donor: ckpt_L0.2-P0.5-S0.5	Relaxation time λ [s] Polymer viscosity μ_p [Pa·s]	0.7000 0.5000	**0.7089** **0.5019**	**+1.27%** **+0.38%**	**0.330%**
Case 3	**Nonlinear Giesekus Model (Blind guess $\alpha_0 = 0.25$)**	Ex-Novo (From Scratch) 40k Adam + 10k L-BFGS Donor: None	Mobility parameter α [-] Relaxation time λ [s] Polymer viscosity μ_p [Pa·s]	0.3500 0.1000 0.5000	**0.3298** **0.1032** **0.5307**	**-5.76%** **+3.16%** **+6.14%**	**0.460%**
Case 4	**Giesekus Extreme Mobility (Strong shear-thinning $\alpha = 0.50$)**	Transfer Learning (TF) 20k Adam + 5k L-BFGS (-50% budget) Donor: ckpt_giesekus_L0.1-A0.35	Mobility parameter α [-] Relaxation time λ [s] Polymer viscosity μ_p [Pa·s]	0.5000 0.3000 0.0500	**0.4665** **0.3077** **0.0522**	**-6.70%** **+2.57%** **+4.40%**	**0.520%**

Grid Convergence Study — Accuracy vs. Mesh Resolution

SETUP: Mesh Robustness | MESH: All | PARAMS: λ =0.1 s, ηp = 0.5 Pa s



Validation and Advantages of Transfer Learning

01. MOTIVATION

Why Transfer Learning?

Training a viscoelastic PINN *from scratch* for each new set of physical parameters or operating conditions is:

- **Computationally expensive** in terms of time and resources.
- **Subject to convergence** towards unfavorable local minima.

02. STRATEGY

Implemented Workflow

1. Pre-training

Training on a base case (low λ)

2. Fine-Tuning

Weight transfer towards severe conditions (higher viscoelasticity or kinematic variations).

03. RESULTS

Advantages and Impact

- **Convergence Acceleration**
- **Numerical Stability**
- **High-Efficiency Scalability**

Phase 2 Inverse Problem Limits

01. IDENTIFIABILITY ISSUE

Identifiability Study on η_s

Our current setup renders velocity fields insensitive to variations in solvent viscosity (η_s):

Velocity Field Invariance

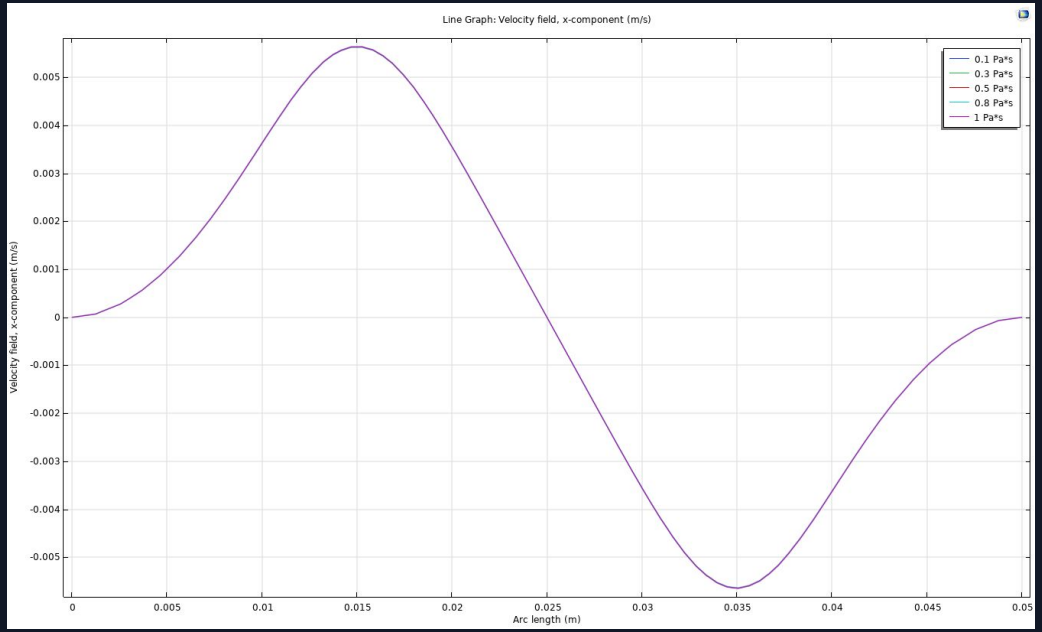
- Changes in η_s produce indistinguishable effects on the velocity profiles.

Optimization Bottleneck

- Loss functions struggle to compute gradients for parameter recovery.

02. CONVERGENCE ANALYSIS

Convergence Plot (η_s)



PTT Inverse Problem Benchmark Results

Benchmark / Dataset	εtrue → εest (ε Error)	λtrue → λest [s] (λ Error)	μp,true → μp,est [Pa·s] (μp Error)	L2(u,v) Kinematics Error
PTT (λ=1.0, ε=0.3) L1-P0.5-S0.5-A0-E0.3	0.300 → 1.7×10−4 **-99.94%**	1.000 → 0.4211 **-57.89%**	0.500 → 0.2128 **-57.44%**	**0.79%**
PTT (λ=0.7, ε=0.5) L0.7-P0.95-S0.05-A0-E0.5	0.500 → 4.4×10−4 **-99.91%**	0.700 → 0.2700 **-61.43%**	0.950 → 0.3687 **-61.19%**	**0.88%**

Key Observation:

While individual parameters **λ, μp, and ε** show high estimation errors (systematic underestimation), the **L2 kinematic error remains extremely low (< 0.9%)**, pointing directly to a parameter identifiability issue.

Proposals to Overcome Setup Limitations

01. SOLVENT VISCOSITY (η_s)

Higher Reynolds Number Regime

Explore a solution space where **inertia becomes more relevant** (higher Re).

Objective:

Amplify the physical impact of solvent viscosity on velocity profiles to break sensitivity invariance.

Expected Outcome:

Sharpen loss function gradients for accurate recovery of η_s.

02. VISCOELASTIC MODEL (PTT)

Exponential PTT Formulation

Transition to an **exponential Phan-Thien-Tanner (PTT)** model formulation.

Objective:

Overcome mathematical & physical limits tied strictly to the stationary regime setup.

Expected Outcome:

Resolve parameter coupling between η_s, η_p, and λ for unique parameter identifiability.

Future Development

01. KINEMATICS MODIFICATION

Flow Field Alteration

Focus on changing the flow field configurations to study non-standard velocity dynamics.

02. STOCHASTIC PERTURBATION

Gaussian Noise Addition

Incorporate Gaussian noise into the synthetic datasets and flow fields to evaluate numerical robustness, regularization stability, and solver resilience under uncertainty.

03. CONSTITUTIVE FORMULATION

Exponential PTT Integration

Transition to exponential PTT (ePTT) instead of linear PTT in ViscoelasticNet.

04. NUMERICAL BOUNDS

Convergence Range Exploration

Conduct thorough parameter sweeps to map out and expand the convergence envelope across varying Weissenberg numbers, Reynolds regimes, and loss weighting setups.

Latest Literature Development

RECENT PUBLICATION (2026)

Simulation of Steady Viscoelastic Creeping Flows with Physics-Informed Neural Networks

José X. Matosa, C. Fernandesa, M. A. Alvesa,*

a CEFT, ALiCE, Faculty of Engineering, University of Porto, Rua Dr. Roberto Frias, s/n 4200-465 Porto, Portugal.

Thank You for Your Attention

Governing Equations & Viscoelastic Formulation

Coupling mass conservation, Cauchy momentum balance, and constitutive stress tensor relations

SYSTEM OF GOVERNING PARTIAL DIFFERENTIAL EQUATIONS

Mass Conservation

$$\nabla \cdot u = 0$$

Incompressible flow condition

Cauchy Momentum Balance

$$\rho(u \cdot \nabla)u = -\nabla p + \nabla \cdot \tau$$

Steady-state momentum transport

Constitutive Relation

$$\tau = \tau s + \tau p$$

Solvent & polymer stress split

DIMENSIONLESS PARAMETERS & REGIMES

Re — Reynolds Number

Characterizes the ratio of inertial forces to viscous forces within the fluid domain, governing flow stability.

Wi — Weissenberg Number

Quantifies fluid elasticity and polymer relaxation time relative to the characteristic rate of flow deformation.

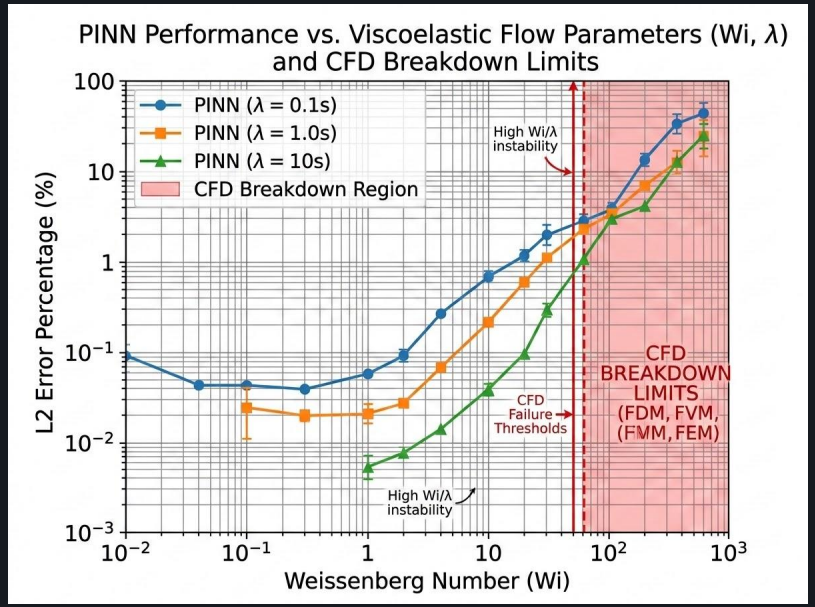
PHYSICAL DOMAIN & RHEOLOGICAL MODEL

Flow Specification: The physical domain models steady-state, incompressible viscoelastic flow under isothermal conditions.

Stress Response: Defined by a viscoelastic constitutive model partitioning the total stress tensor into Newtonian solvent and elastic polymer contributions.

System Coupling: Momentum balance and non-linear constitutive equations dictate the final flow behavior across the domain.

High Weissenberg Number Problem Mitigation



Quantitative Benchmark: L2 Error vs. Weissenberg Number (Wi)

Comparative breakdown highlighting CFD failure thresholds against mesh-free PINN stability.

Regime Stability & Computational Overhead

Bypassing Numerical Breakdown

The physics-informed framework demonstrates robust performance in complex regimes, effectively bypassing the High Weissenberg Number Problem.

By eliminating complex meshing requirements for viscoelastic flow domains, the model maintains numerical stability and accuracy even in highly non-linear regimes.

High-Fidelity Prediction Capability

Non-Linear Stress & Velocity Kinematics

The mesh-free solver achieves high-fidelity predictions of both velocity kinematics and non-linear stress tensors.

It mitigates the numerical breakdown and steep computational overheads traditionally associated with high elasticity and complex relaxation time behaviors.

Non so se aggiungere anche questo forse alla fine

COMPUTATIONAL ENGINEERING IMPLICATIONS

Take-Home Messages & Method Positioning

Core Strengths

- Ideal for solving inverse problems and parameter identification.
- Seamless sparse experimental data assimilation (e.g., PIV velocity fields).
- Quantifies unmeasurable latent physical variables directly.

Practical CFD Positioning

- Does not aim to replace mature forward CFD solvers (FVM / FEM) for high-fidelity forward predictions.
- Acts as an integrative framework, bridging real experimental data with physical models.

Experimental Data (PIV) + Physical Laws → PINN Integration